

Nikita Kazeev

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Research Scientist focusing on Reliability and Uncertainty in Agentic Systems

WORK EXPERIENCE

PERFOMAX <i>AI Advisor</i>	2026–present
CONSTRUCTOR GROUP <i>Machine Learning Consultant</i>	2021–present
NATIONAL UNIVERSITY OF SINGAPORE <i>Postdoc under Kostya Novoselov</i>	2022–present

Leadership

- Program Chair for the [AI4X 2025/6 conferences](#)
- Main organiser of the [ICLR 2025 Workshop on Machine Learning Multiscale Processes](#)
- Co-PI of a US\$3.4m AI Singapore grant on multiscale machine learning

LLM Agent Safety & Alignment [ICML 2026 Workshop](#) · [ICML 2026 Workshop](#) · [GitHub](#)

- Conceived and developed a novel geometric framework to predict multi-step reasoning failures in LLM agents using semantic abstraction trajectories
- Designed uncertainty quantification (UQ) algorithms for LLM reasoning loops, improving agent error detection and calibration
- Developed and open-sourced [reference-checker](#), a tool for verifying claims and citation references in scientific papers to detect and mitigate LLM hallucinations
- Authored a position paper on Pluralistic Alignment, advocating for aligning AI models with aspirational human values rather than replicating behavioral flaws

Wyckoff Transformer, a generative model for materials [ICML 2025](#) · [GitHub](#)

- Alignment by design: enforced physical principles directly at the architectural level as hard constraints, mathematically guaranteeing plausible generations
- Solved a systemic failure mode where generative models default to asymmetric outputs despite symmetric training data, mitigating this structural bias via coordinate-free representations
- Increased materials discovery yield (S.U.N.) by 1.24x while cutting inference time by 4 orders of magnitude (to 0.05 GPU-ms per structure)
- Orchestrated large-scale (10k structures) DFT computation loops to verify out-of-distribution stability; led a team of 6

(CERN | YANDEX → HSE UNIVERSITY) *Researcher* 2014–2022
[2025 Breakthrough Prize in Fundamental Physics](#) as a member of LHCb.

Generative models uncertainty estimation [paper](#) · [conference](#) · [code 1](#) · [code 2](#)

- Developed the first methods for estimating uncertainty of conditional GANs
- Led a team of 3 students

Machine learning on noisy data [paper 1](#) · [paper 2](#) · [code](#)

- Derived a statistically rigorous method to train classifiers on sPlot-weighted (label-noisy) data, standard in high-energy physics

CatBoost: Combatting Biases in Boosting [paper](#)

- Designed and built distributed experiment infrastructure to scale systematic bias analysis across large-scale tabular datasets
- Investigated dynamic boosting formulations to eliminate target leakage and prediction shift, mitigating statistical bias in gradient boosting

EDUCATION

HSE University & Sapienza Università di Roma <i>PhD in Computer Science and Physics (Double Degree)</i> Supervisors: Andrey Ustyuzhanin & Barbara Sciascia Thesis: Machine Learning for particle identification in the LHCb detector	2016–2020
Product Management course at Yandex <i>Focused coursework in product strategy and execution.</i>	2018
Yandex School of Data Analysis <i>Master's-level CS programme</i> Algorithms, machine learning, deep learning, and distributed systems.	2013–2015
Moscow Institute of Physics and Technology <i>MS in Physics</i> Optimisation of data processing of the LHCb experiment.	2014–2016
<i>BS in Physics</i> Wave-packet molecular dynamics of electrons in nonideal plasma.	2010–2014
International Junior Science Olympiad (IJSO) Silver medal	Korea, 2008

MENTORSHIP

- Mentored 9 students ([1](#), [2](#), [3](#), [4](#), [5](#), [6](#), [7](#), [8](#), [9](#)), 3 interns, and student workshop projects ([1](#), [2](#)).

TEACHING & OUTREACH

- Pitch research to the general public through major newspaper coverage ([1](#), [2](#)), blog posts ([1](#), [2](#)), a [science slam](#), interviews ([1](#), [2](#)), and a public [talk](#).
- Lecturer in the [award-winning](#) Coursera Advanced Machine Learning [specialisation](#)
- Delivered more than 20 [conference talks](#).

SERVICE

- Reviewer for NeurIPS, RSC Advances, Machine Learning: Science and Technology, Digital Discovery
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

SKILLS

- **Agent Reliability & UQ** — developed geometric and statistical frameworks to predict LLM agent reasoning failures using semantic abstraction trajectories.
- **Aspirational Alignment & Verification** — formulated alignment strategies based on high-level human values and established benchmarks (e.g. Foresight-Phys) for validating AI systems in complex environments.
- **Uncertainty quantification** — built the first methods for estimating the uncertainty of conditional GANs; calibrated generative and discriminative models.
- **Inference and systems at scale** — C++ production integration, large-scale DFT/VASP on HPC clusters, and reproducible experiment pipelines.
- **Research leadership** — lead multidisciplinary teams, chair conferences, and mentor students from idea to publication.

ML & AI • Python • PyTorch • C++ • HPC • Physics • Research Writing & Speaking • Leadership & Mentorship

PRESENTING AT ICML 2026

July 11, 12:15–13:30 — Nikita Kazeev and Ian Babich. [“Foresight-Phys: A Benchmark for Forecasting the Results of Physical Experiments”](#). In: *Forecasting as a New Frontier of Intelligence Workshop at ICML 2026*.

July 11, 15:00–16:30 — Nikita Kazeev and Phan Bui Nhat Huyen. [“Position: Align AI to Our Aspirations, Not Our Flaws”](#). In: *Pluralistic Alignment Workshop at ICML 2026*.

July 11, 15:30–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. [“The Geometry of Reasoning Failure: Predicting Agent Errors from Semantic Scale Trajectories”](#). In: *Statistical Frameworks for Uncertainty in Agentic Systems Workshop at ICML 2026*.

July 10, 11:40–13:30 & 16:10–17:00 — Nikita Kazeev and Andrey Ustyuzhanin. [“Uncertainty Quantification for LLM Agents via Semantic Abstraction Trajectories”](#). In: *Structured Probabilistic Inference & Generative Modeling Workshop at ICML 2026*.

SELECTED PUBLICATIONS

Nikita Kazeev, Wei Nong, Ignat Romanov, Ruiming Zhu, Andrey E Ustyuzhanin, Shuya Yamazaki, and Kedar Hipalgaoonkar. [“Wyckoff Transformer: Generation of Symmetric Crystals”](#). In: *Proceedings of the 42nd International Conference on Machine Learning*. Ed. by Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu. Vol. 267. Proceedings of Machine Learning Research. PMLR, 13–19 Jul 2025, pp. 29495–29526.

Maxim Borisyak and Nikita Kazeev. [“Machine Learning on data with sPlot background subtraction”](#). In: *Journal of Instrumentation* 14.08 (2019). Alphabetic order, I’m the corresponding author.

Denis Derkach, Nikita Kazeev, Fedor Ratnikov, Andrey Ustyuzhanin, and Alexandra Volokhova. [“Cherenkov detectors fast simulation using neural networks”](#). In: *Nuclear Instruments and Methods in Physics Research Section A* (2019). Alphabetic order, I’m the corresponding author.